

Sub-clinical neurobehavioral abnormalities associated with low level of mercury exposure through fish consumption.

In order to assess early neurotoxic effects associated with relatively low levels of mercury absorbed through fish eating, two groups of 22 adult male subjects, habitual consumers of tuna fish, and 22 controls were examined using a cross-sectional field study. The assessment included neurobehavioral tests of vigilance and psychomotor function, hand tremor measurements and serum prolactin assessment. Mercury in urine (U-Hg) and serum prolactin (sPRL) were measured in all exposed subjects and controls, whereas measurements of the organic component of mercury in blood (O-Hg) were available for only 10 exposed and six controls. U-Hg was significantly higher among exposed subjects (median 6.5 microg/g of creatinine, range 1.8-21.5) than controls (median 1.5 microg/g of creatinine, range 0.5-5.3). The median values of O-Hg were 41.5 microg/l among the tuna fish eaters and 2.6 microg/l in the control group. Both U-Hg and O-Hg were significantly correlated with the quantity of fish consumed per week. Significant differences in sPRL were found between exposed (12.6 ng/ml) and controls (9.1 ng/ml). Individual sPRL were significantly correlated with both U-Hg and O-Hg levels. The neurobehavioral performance of subjects who consumed tuna fish regularly was significantly worse on color word reaction time, digit symbol reaction time and finger tapping speed (FT). After considering the education level and other covariates, the multiple stepwise regression analysis indicated that O-Hg concentration was most significantly associated with individual performance on these tests, accounting for about 65% of the variance in test scores.